INCH-POUND

MIL-PRF-1/263G
22 February 2008
SUPERSEDING
MIL-E-1/263F
17 February 1981

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING
TYPE 6X4WA

Inactive for new design
after 7 March 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Rectifier, miniature, full-wave, vacuum.

Outline - - - 5-3 (EIA).

Base - - - E7-1.

Envelope - - - T5-1/2.

Cathode - - - Coated unipotential.

Base connections:

Pin No.	- - -	1	2	3	4	5	6	7
Element	- - -	2a	nc	h	h	nc	1a	k

ABSOLUTE RATINGS:

Parameter:	Ef	Epp/a	Ehk	Rl	CL	Io	ib/a	i surge/a	TE	Alt
Unit:	V	V ac	v	Ohms	µF	mA dc	ma	a	°C	ft
Maximum:	6.9	See note 1	+100	---	---	See note 1	210	1.5	165	See note 1
Minimum:	5.7	---	-450	---	---	---	---	---	---	---
Test conditions:	6.3	400 (See note 2)	---	8,200	8	---	---	---	---	---

GENERAL:

Qualification - Required.

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TABLE I. Testing and Inspection.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 11)	Symbol	Limits		Unit
					Min	Max	
	Qualification inspection						
1031	Variable-frequency vibration	No voltages applied	---	---	---	---	---
	Conformance inspection, part 1						
1201	Short and discontinuity detection		0.4	---	---	---	---
1231	Emission (1)	Eb = 50 V dc (see note 3)	0.4	Is	100	---	mA dc
1301	Heater current		0.4	If	550	650	mA
1353	Operation of rectifiers	See notes 4 and 5	0.4	Io	51	---	mA dc
1336	Heater-cathode leakage	Ehk = - 450 V dc	0.4	Ihk	---	75	µA dc
	Conformance inspection, part 2						
1211	Insulation of electrodes	See note 3	2.5	R	10	---	MegΩ
1002	Barometric pressure, reduced	Pressure = 140 ±5 mmHg; voltage = 980 V ac	6.5	---	---	---	---
1031	High-frequency vibration	No voltages applied	6.5	---	---	---	---
1231	Emission (2)	Ef = 5.5 V; Eb = 50 V dc (see note 3)	2.5	Is	90	---	mA dc
1041	Shock	450 G; Epp/a = 0	6.5	---	---	---	---
1031	Vibration-fatigue		6.5	---	---	---	---
---	Post-shock and vibration-fatigue test end points:						
1336	Heater-cathode leakage		---	Ihk	---	150	µA dc
1353	Operation of rectifiers		---	Io	48	---	mA dc
1121	Base strain	See note 7	---	---	---	---	---
2126	Envelope strain		2.5	---	---	---	---
1105	Permanence of marking		---	---	---	---	---

See notes at end of table.

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TABLE I. Testing and Inspection. -Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Symbol	Limits		Unit
				Min	Max	
	Conformance inspection, part 3					
1506	Heater-cycling life	Ef = 7.5 V; Ehk = 100 V dc; Epp/a = 0; Rl = CL = 0 (1 minute "on", 4 minutes "off")	---	---	---	---
---	Heater-cycling life-test end point:					
1336	Heater-cathode leakage		Ihk	---	150	µA dc
1501	Intermittent life	TA = room See note 8 TE = 165°C (min) (see notes 9 and 10)	---	---	---	---
---	Intermittent life-test end points (500 hours):	See note 11				
1353	Change in operation of rectifiers of individual tubes		ΔIot	---	8.5	%
1301	Heater current		If	550	660	mA
1336	Heater-cathode leakage		Ihk	---	90	µA dc
---	Intermittent life-test end points (1,000 hours):	See note 11				
1353	Change in operation of rectifiers of individual tubes		ΔIot	---	11.5	%
1301	Heater current		If	550	666	mA
1336	Heater-cathode leakage		Ihk	---	90	µA dc

NOTES:

1. To simplify the application of the maximum ratings to circuit design, the absolute-maximum ratings are presented in chart form as rating charts I, II, III, and IV. Operating points should be so selected that the boundary limits on rating charts, I, II, III, and IV are not exceeded with a bogey tube under the worst probable operating conditions with respect to supply-voltage variation, equipment component variation, equipment control adjustment, and environmental conditions.

   A brief description of each of the rating charts is given below. The values of ac supply voltage as presented refer to the unloaded supply voltages per anode.

   a. Rating chart I (see figure 1). This chart presents the maximum ratings for ac anode supply voltage and dc output current. The boundary FAEDG defines the limits for capacitor-input filter operation and the boundary FABCDG defines the limits for choke-input filter operation.

   b. Rating chart II (see figure 2). This chart provides a convenient method for checking conformance with the maximum steady-state peak-anode-current rating. Rating chart II applies to capacitor-input filter operation only.

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TABLE I. Testing and Inspection. -Continued.

c. Rating chart III (see figure 3). This chart shows the minimum value of anode supply resistance (Rs) required to remain within the transient peak-anode-current rating. The value of Rs should be such that it lies to the left of the line on the rating chart at the highest probable value of line voltage. Rating chart III applies to capacitor-input filter operation only.

d. Rating chart IV (see figure 4). This chart presents the maximum ratings for ac anode supply voltage and altitude. Rating chart IV refers to both capacitor-input filter and choke-input filter operation.

2. All values of Epp/a refer to the unloaded supply voltage. The ratings refer to rectifier operation with sinusoidal supply voltages within the frequency range of 25 to 1,000 Hz.

3. Test each section separately.

4. In a full-wave circuit, adjust Zp/a so that a bogey tube gives Io = 55 mA dc. A bogey tube has a tube drop of Etd = 22 V dc at Is = 50 mA dc per anode. Ehk = Eo.

5. This test shall be performed at the conclusion of the holding period.

6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 12 month rule shall apply.

7. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (MIL-STD-1311, method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

   a. Accepted if no defective for class "A", "B", or "C" defects, respectively (see MIL-STD-1311, method 1121), or if no defectives are found in the sample.

   b. Rejected if any defectives for class "A", "B", or "C" defects, respectively, or if any defectives are found in the sample.

8. In a full-wave life-test circuit, the values specified for R1 and CL may be considered as approximate and shall be adjusted initially to give not less than Io = 55 mA dc and ib/a = 200 ma with a bogey tube (see note 4). Ehk = Eo.

9. The life-test sample shall consist of 20 tubes per lot and no failures shall be permitted. In the event of rejection of the first sample, due to failure, a second sample of the lesser of 20 tubes or 10 percent of lot size shall be selected from the lot. Acceptance shall then be based on the second sample and no tube failures shall be permitted.

10. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (MIL-STD-1311, method 1226), will be satisfied if a tube having bogey Etd (±10 percent at Is = 50 mA dc per anode) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

11. This specification sheet utilizes an accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

12. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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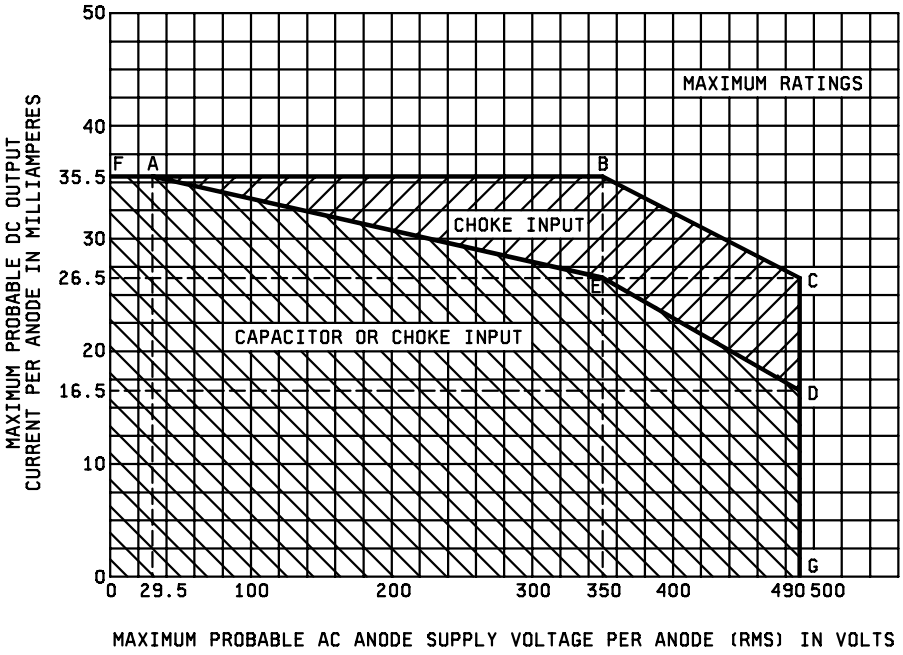


FIGURE 1. Rating chart I.

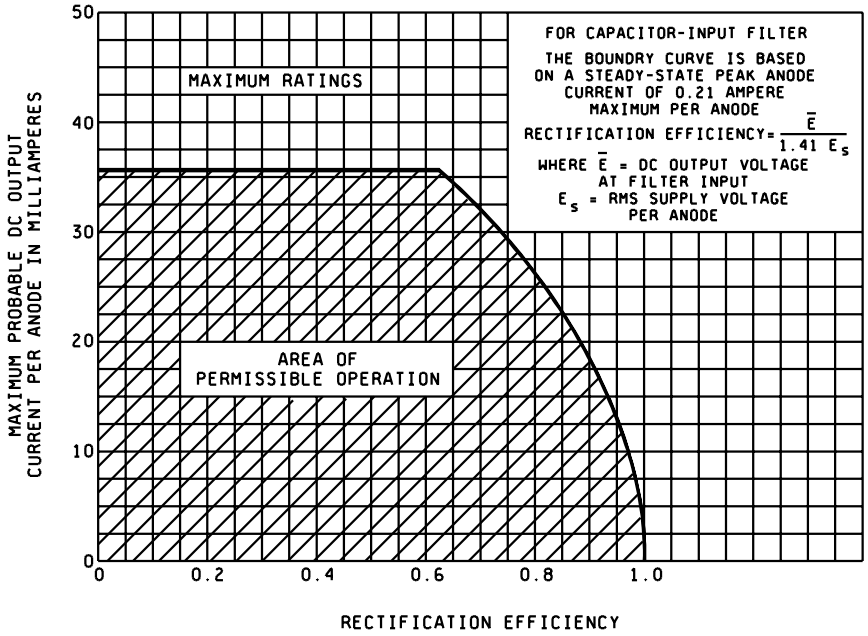


FIGURE 2. Rating chart II.

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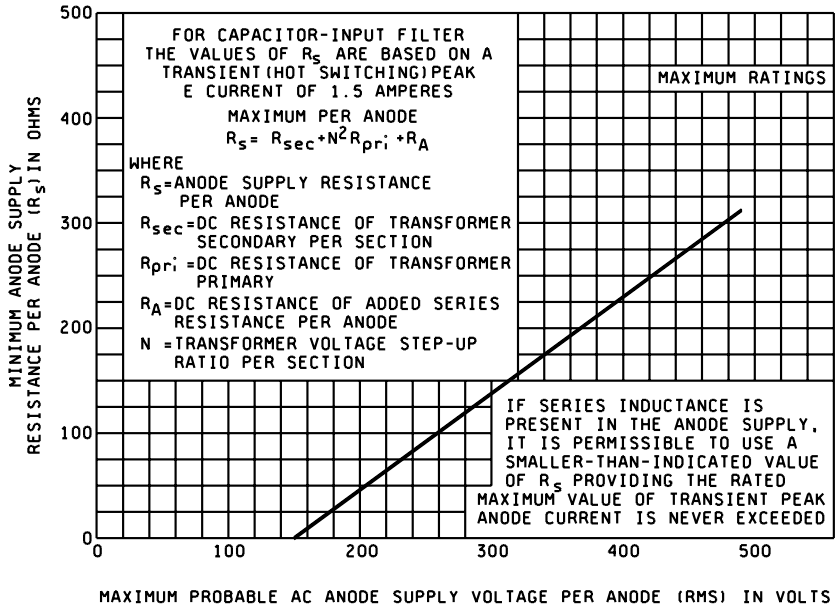


FIGURE 3. Rating chart III.

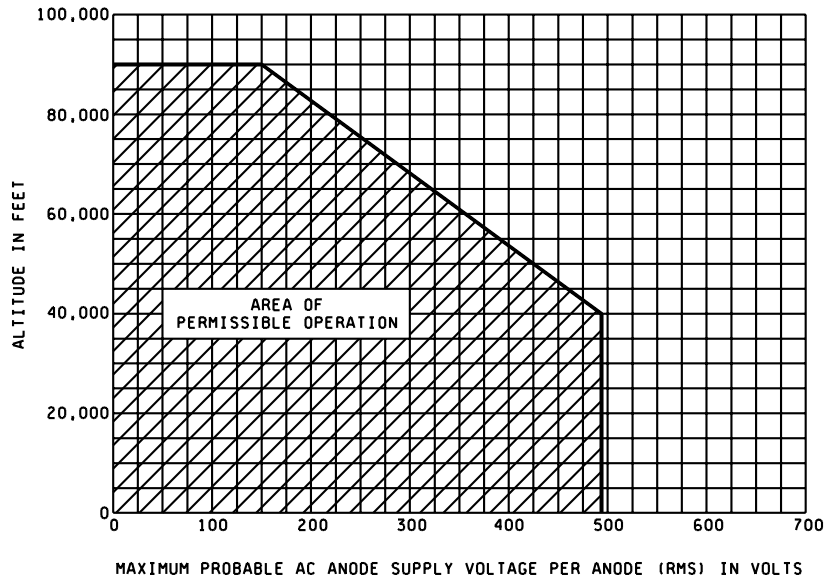


FIGURE 4. Rating chart IV.

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Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311.

Changes from previous revision. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2008-014)

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force - 99

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